

Predicting Forest Cover Type

Group 6 Team Members: Jane Sanchez, Julie Huynh, Jingda Li, Jessy Donelle, Jeremy May

1. Introduction:

As environmental conditions continue to change at an unprecedented rate, forest managers require smarter and more efficient approaches to monitor and manage natural resources. Using data-driven spatial modeling has become fundamental for modern ecological stewardship. As such, predicting forest cover type is highly valuable since it allows researchers and land managers to identify vegetation structures across large expanses without conducting costly, time-intensive field surveys.

The following report examines a Forest Cover Type dataset, that was originally sourced from the US Forest Service (USFS) and hosted by the UCI Machine Learning Repository, to evaluate how topographic and environmental indicators dictate forest cover types across the Roosevelt National Forest (Blackard, 1998). The dataset used in this project was obtained from Kaggle (UCI Machine Learning, n.d.).

This dataset was chosen because it is large enough to build robust, meaningful predictive models without being so unwieldy or complex that resources are drained by data cleaning. Furthermore, this dataset presents an ideal benchmark to compare various machine learning approaches on a complex multi-class classification problem, ultimately supporting downstream applications in forest management, conservation efforts, wildfire planning, and environmental monitoring.

2. Objective:

What are you setting out to prove or predict?

The primary objective of this project was to **develop and compare multiple supervised machine learning models to determine which algorithm most accurately predicts forest cover type from environmental and topographic variables**. Using a Forest Cover Type dataset, five classification algorithms—Logistic Regression, Linear Support Vector Classifier (Linear SVC), Decision Tree, Random Forest, and Neural Network—were trained and evaluated on the same dataset to identify the model that provided the best overall classification performance.

In addition to identifying the highest-performing model, the project sought to better understand the strengths and limitations of linear and nonlinear classification techniques, the effect of class imbalance on model performance, and the importance of environmental variables in distinguishing different forest cover types.

What is the business or other problem that you are trying to address with this model?

Accurately identifying forest cover type is an important task in forestry, environmental management, and land-use planning. Traditional forest inventories rely heavily on field surveys, which require significant time, labour, and financial resources to collect vegetation information across large or remote landscapes (Bonham, 2013). As forest ecosystems continue to face increasing pressures from climate change, wildfires, invasive species, and human development, there is growing demand for faster and more scalable methods of monitoring forest conditions (USDA Forest Service, 2020).

Machine learning offers an efficient alternative by using readily available environmental and cartographic variables to predict forest cover type without requiring extensive field sampling. The Forest Cover Type

dataset contains topographic and ecological variables—including elevation, slope, aspect, hillshade, soil type, and wilderness area—that are known to influence vegetation distribution and species composition (Blackard & Dean, 1999). Developing classification models from these variables can support forest managers in producing accurate vegetation maps while reducing the cost and effort associated with traditional survey methods.

This analysis has practical applications in forest inventory management, biodiversity conservation, wildfire risk assessment, habitat monitoring, ecosystem management, and environmental planning (FAO, 2020; USDA Forest Service, 2020). By identifying the algorithm that most accurately predicts forest cover type, this project provides evidence to support the selection of machine learning methods for future ecological mapping and decision-support systems.

3. Data Preparation:

Data Source (What was your data source (e.g., a standard machine learning data set, open data, etc.)?):

The data source for this report is a standard, publicly available machine learning dataset—specifically, the Forest Cover Type dataset hosted by the UCI Machine Learning Repository (Blackard, 1998). Originally derived from data compiled by the US Forest Service (USFS) Region 2 Resource Information System (RIS), this dataset contains 581,012 observations (representing 30x30 meter spatial cells) across four wilderness areas in the Roosevelt National Forest of Colorado. The dataset comprises 54 total attributes, combining continuous quantitative variables (such as elevation, slope, aspect, hillshade indices, and distances to hydrological features and roadways) with binary categorical variables (representing multiple wilderness designations and 40 distinct soil types).

Data Quality (How good was the data quality?): The data quality for the Forest Cover Type dataset is exceptionally high and clean. The dataset contains complete records with no missing values across any of the attributes, which eliminates the need for data imputation strategies or removing incomplete rows. The data is logically organized and standardized into continuous quantitative metrics (such as elevation, slope, and distances) and binary categorical indicators (such as soil types and wilderness designations).

Data Procurement (What did you need to do to procure it?): Procuring the dataset required accessing the UCI Machine Learning Repository online archive, navigating to the Forest Cover Type dataset page, and downloading the raw data files. The downloaded files were imported into a development environment (i.e. Google Colab) using Python's pandas library to prepare for exploratory data analysis and modeling.

Data Preparation (What tools or code did you need to use to prepare it for analysis?): To prepare the data for model training, the following tools and libraries were utilized: Pandas, NumPy, and Scikit-Learn. The data itself was already clean so no imputation was required and binary features like soil type were already encoded. Continuous values were standardized for better model performance.

What challenges did you face? One of the challenges faced with this data set is that there was class imbalance. The distribution of the seven forest cover types is heavily skewed. Classes 1 and 2 (1: 'Spruce/Fir', 2: 'Lodgepole Pine') make up the vast majority of the dataset, whereas classes 4 and 5 (4: 'Cottonwood/Willow', 5: 'Aspen') are underrepresented. By plotting a histogram of the frequency of each Cover Type, this observation was noted and the balanced class_weight hyperparameter was utilized in training the various models.

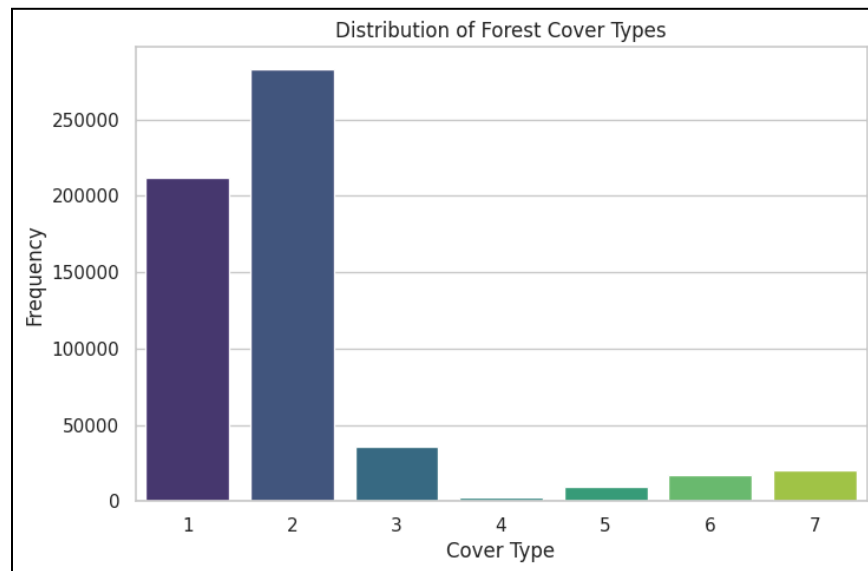


Figure 1. Distribution of Forest Cover Types

Class overlap was also evident within the dataset, with several forest cover types occupying similar regions of the feature space. The LDA plot (Annex A, Figure A.2) shows that some classes remain mixed even in a label-guided linear projection, while the t-SNE plot (Annex A, Figure A.1) highlights local clusters and overlapping transition regions. Together, these plots show that the classes are only partially separable rather than clearly distinct.

Another challenge was the size of the dataset, which contains over 581,000 records and increased the computational cost of model training. Most models were trained using the full dataset; however, the RBF SVC required a random subsample of 50,000 records to make training computationally feasible. To further reduce unnecessary computation, model training results were persisted so they did not need to be recalculated in each session.

The last challenge faced in this data set is feature overlap and non-linearity. Environmental and topographic variables such as elevation, aspect and slope, interact in complex, non-linear ways. Simple linear models may not capture these complex relationships accurately and tree-based models may achieve stronger performance, which are explored in the sections below.

4. Model Design:

Describe your (two or more) models:

Five supervised machine learning algorithms were developed and compared: Logistic Regression, Linear Support Vector Classifier (Linear SVC), Decision Tree, Random Forest, and Neural Network. These models were selected because they represent a diverse range of linear, nonlinear, tree-based, ensemble, and deep learning approaches, allowing their predictive performance to be evaluated on the same multiclass classification problem.

Before model training, the dataset was divided into 80% training data and 20% testing data using a stratified train-test split (`random_state = 42`). Stratification preserved the original proportions of all seven

forest cover types in both datasets, ensuring that the class imbalance present in the original dataset was maintained during both training and evaluation. Using the same training and testing datasets for every model ensured that differences in predictive performance could be attributed to the algorithms themselves rather than differences in the underlying data.

Continuous predictor variables (e.g., elevation, slope, hillshade, and distance measures) were standardized using StandardScaler for algorithms that are sensitive to feature scale, including Logistic Regression, Support Vector Machines, and the Neural Network. Binary indicator variables representing wilderness areas and soil types remained unchanged because they were already encoded as 0/1 variables. Tree-based models (Decision Tree and Random Forest) were trained using the original feature values because these algorithms are unaffected by variable scaling.

To address the class imbalance present in the dataset, most models incorporated balanced class weighting, assigning greater importance to underrepresented forest cover types during training. This approach encouraged the models to learn minority classes more effectively while reducing bias toward the dominant cover types. Model performance was evaluated using an independent test set and compared primarily using Macro F1-score, which assigns equal importance to each forest cover type regardless of class frequency.

How did you choose hyperparameters if they were required?

The most impactful parameters for each model were chosen along with the common exploratory values for each, and were selected using a combination of default settings and model-specific tuning procedures. For computationally intensive algorithms, cross-validation was used to identify high-performing parameter combinations while minimizing unnecessary computation. RandomizedSearchCV was used for the Decision Tree and Random Forest models because of their relatively large hyperparameter spaces, while GridSearchCV was used for the RBF Support Vector Machine to exhaustively evaluate combinations of the regularization parameter and kernel parameter.

For the decision tree, the most impactful hyperparameter was max_depth, which limits how deep the decision tree can grow, which influences the model's ability to learn patterns and generalize unseen data. Additionally, finding the optimal max_depth allowed for any overfitting to also be mitigated.

Unlike the classical models, the neural networks were tuned iteratively based on validation performance rather than using a predefined hyperparameter search. The neural network was initially configured as a small ReLU network (128–64 neurons), with network width and depth subsequently increased and residual blocks introduced when validation results indicated underfitting. The final learning rate (0.001), batch size (1024), dropout (0.2–0.3), early stopping, and ReduceLROnPlateau settings were selected based on validation macro F1 and training stability. Because Keras does not provide a class_weight='balanced' option directly, class weights were calculated from the training data and supplied to the model during training to account for class imbalance and improve performance on minority classes.

Macro F1 was used as the primary evaluation metric because the forest cover classes are imbalanced. It evaluates the F1 score of each class separately and gives each class equal importance when calculating the

overall score. This provides a more balanced measure of model performance than accuracy alone, particularly when a model performs well on common classes but poorly on less frequent ones.

Why did you choose those particular machine learning models?

The selected models represent a diverse range of supervised learning approaches with varying levels of complexity, flexibility, and interpretability. Logistic Regression and Linear SVC were included as linear baseline models to determine how well forest cover types could be separated using linear decision boundaries. RBF Support Vector Classifier (RBF SVC) extends this approach by introducing a nonlinear kernel capable of modelling more complex decision boundaries. Decision Trees provide an interpretable nonlinear model that can naturally capture interactions among environmental variables, while Random Forests build upon this approach through ensemble learning to improve predictive performance and reduce overfitting. Finally, Neural Networks were used because they can learn complex nonlinear patterns and feature interactions that simpler models often miss. The Residual Neural Network was included to determine whether residual connections could improve learning of complex nonlinear relationships and feature interactions. These connections allow information to bypass intermediate layers, which can improve training and enable the network to learn both simpler and more complex patterns within the data.

These models were selected because they employ fundamentally different classification strategies, making them well suited for comparison on a multiclass ecological classification problem. Comparing linear and nonlinear models allows us to determine whether the relationships between environmental variables and forest cover type are primarily linear or whether more sophisticated models are required to capture complex interactions. Furthermore, evaluating these algorithms using common training and testing datasets enables meaningful comparisons of predictive performance, class balance, interpretability, computational efficiency, and robustness to the class imbalance present in the dataset.

5. Model Evaluation:

How well did the model perform on the set-aside testing dataset?

Each machine learning model was evaluated using the same independent 20% testing dataset, ensuring that performance comparisons were fair and unbiased. To assess classification performance, several complementary evaluation metrics were calculated, including Accuracy, Macro Recall, Macro F1-score, and Weighted F1-score. Although overall accuracy provides a measure of the proportion of correctly classified observations, it can be misleading for imbalanced datasets because it is heavily influenced by the majority classes. Therefore, Macro F1-score was selected as the primary performance metric since it calculates the F1-score for each forest cover type independently and assigns equal importance to all seven classes. Macro Recall was also reported to evaluate each model's ability to correctly identify observations from every forest cover type, while Weighted F1-score provides an overall measure of classification performance that accounts for the relative frequency of each class. Together, these metrics provide a comprehensive assessment of model performance across both common and rare forest cover types.

Overall, the results demonstrate substantial differences in predictive performance among the machine learning algorithms. While all models were capable of classifying forest cover type, the nonlinear approaches consistently outperformed the linear models across nearly every evaluation metric. As illustrated in the model comparison heat map (Figure 2), the Random Forest classifier achieved the strongest overall Accuracy (95.7%), Macro Recall (94.7%), Macro F1-score (93.3%), and Weighted

F1-score (95.7%), indicating that it provided the most balanced and accurate classification performance across all seven forest cover types. The heat map also highlights the relative strengths and weaknesses of each algorithm across the evaluation metrics (darker shading indicates stronger performance), making it clear that the nonlinear models consistently outperformed the linear approaches. The Decision Tree and Residual Neural Network also demonstrated excellent predictive performance, whereas the Logistic Regression and Linear SVC models produced considerably lower Macro F1-scores, particularly for the minority forest cover types. These findings suggest that the relationships between environmental variables and forest cover type are highly nonlinear and are more effectively captured by tree-based ensemble methods and neural networks than by linear classification algorithms. Based on its superior performance across all evaluation metrics, excellent classification of both majority and minority forest cover types, and strong ability to model complex nonlinear relationships, the Random Forest classifier was selected as the best-performing model for this study and the recommended approach for predicting forest cover type within this dataset.

Table 1. Model Evaluation Metrics

Model Family	Model	Accuracy	Macro Recall	Macro F1	Weighted F1
Logistic Regression	LogisticRegression class_weight=balanced	0.599012	0.709271	0.506867	0.627815
Linear SVC	LinearSVC class_weight=balanced	0.680938	0.622347	0.542598	0.688286
Decision Tree	DecisionTree class_weight=balanced (tuned)	0.939055	0.903897	0.902864	0.939067
Random Forest	RandomForest class_weight=balanced (tuned)	0.956714	0.946589	0.932808	0.956811
RBF SVC	SVC RBF class_weight=balanced (tuned)	0.790470	0.835140	0.725130	0.797257
Neural Net	Neural Net (residual, balanced class weights)	0.926757	0.958805	0.892076	0.927997

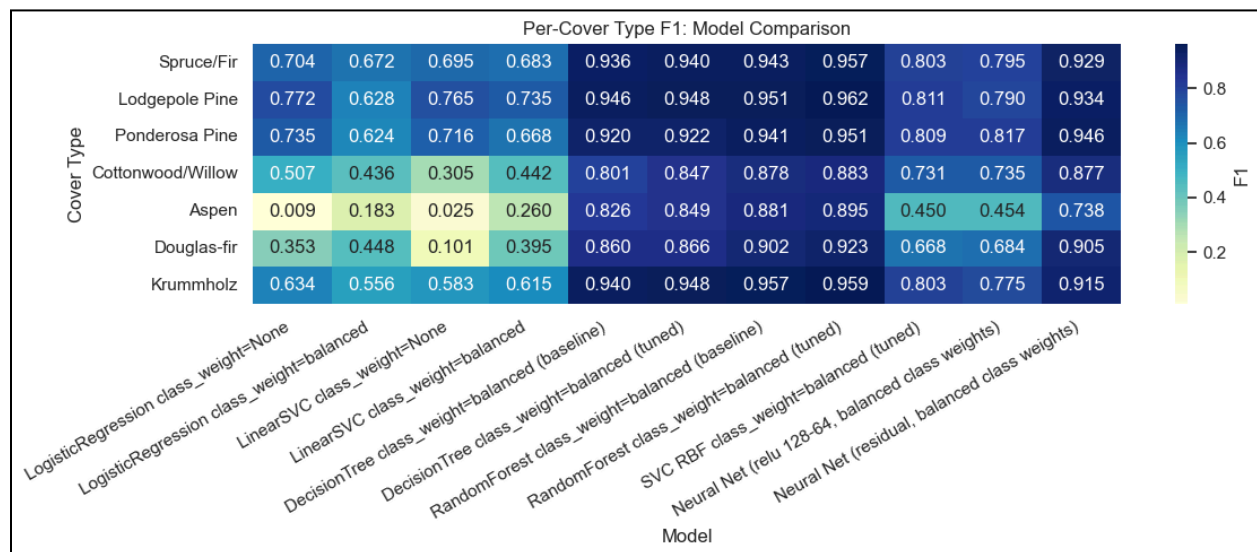


Figure 2. Comparison of machine learning model performance across evaluation metrics.

Linear vs. Non-Linear Models

The results show a clear advantage for the nonlinear models. Logistic Regression and Linear SVC plateaued at relatively low Macro F1 scores, suggesting that additional training data provided little improvement once their linear decision boundaries were established. Their performance was also weakest on less common classes such as Aspen, Cottonwood/Willow, and Douglas-fir. In contrast, the RBF SVC, Decision Tree, Random Forest, and Residual Neural Network achieved much stronger Macro F1 scores and substantially better per-class performance. The learning curves also show continued improvement for the nonlinear models as training size increased. This suggests forest cover classes are better represented by nonlinear decision boundaries and complex feature interactions.

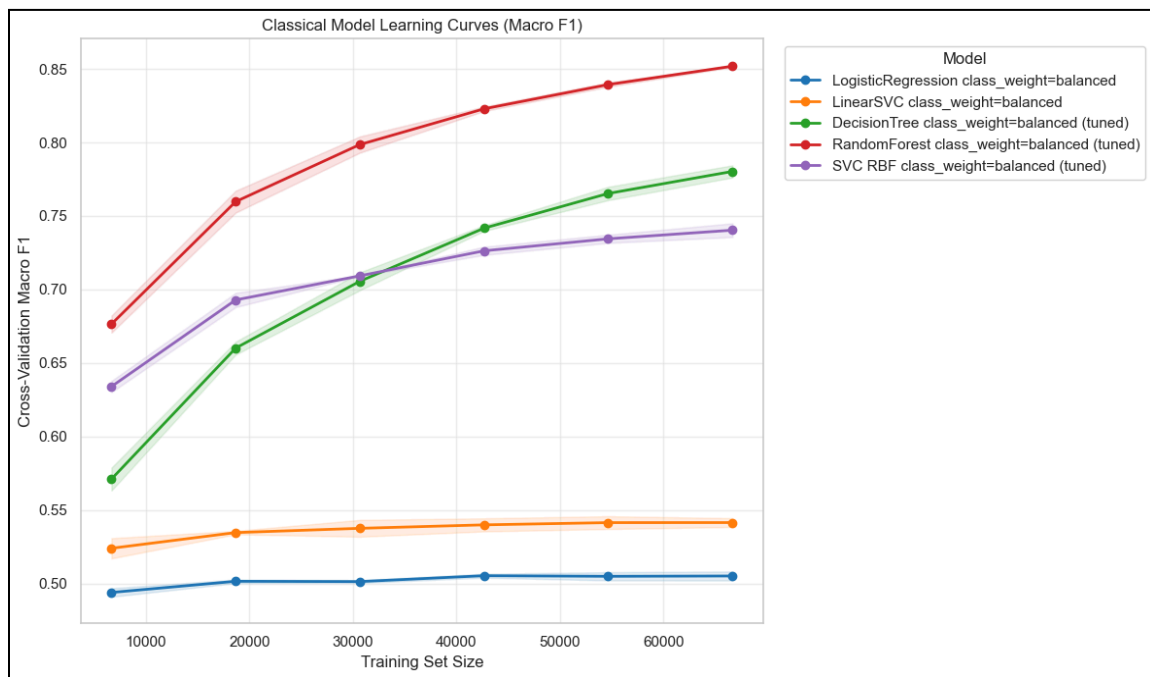


Figure 3. Classification Model Learning Curves

The neural network learning curves were plotted separately from the classical models because they represent performance across training epochs rather than across increasing training-set sizes. This provides a clearer representation of the neural networks' training behaviour and convergence (Annex A, Figure A.3).

If you attempted an ensemble model how did it perform vs. the individual models in the ensemble?

A Random Forest classifier is an ensemble of Decision Tree classifiers. While a standalone Decision Tree provides an intuitive and highly interpretable model by recursively splitting data based on feature thresholds, it is often prone to high variance and overfitting when left unconstrained. Both the Random Forest classifier and Decision Tree classifier were evaluated in this analysis, where the Random Forest classifier performed better in every single evaluation metric. This is likely due to the fact that the Random Forest classifier is exceptional at generalization, robust against overfitting and handles complex non-linear interactions very well.

Table 2. Random Forest vs. Decision Tree Evaluation Metrics

Model	Accuracy	Macro Recall	Macro F1	Weighted F1
RandomForest class_weight=balanced (tuned)	0.956714	0.946589	0.932808	0.956811
RandomForest class_weight=balanced (baseline)	0.944812	0.934233	0.921951	0.944899
DecisionTree class_weight=balanced (tuned)	0.939055	0.903897	0.902864	0.939067
DecisionTree class_weight=balanced (baseline)	0.935286	0.884234	0.889800	0.935199

Based on the Random Forest vs. Decision Tree evaluation metrics presented in Table 2, the baseline and tuned Random Forest models outperformed the tuned Decision Tree model in terms of accuracy, macro recall, macro F1 and weighted F1 scores. The overall accuracy of the tuned Random Forest was 95.67%, beating the Decision Tree at 93.91%. The macro F1 score of the tuned Random Forest was 93.28%, which is about 3% higher than the Decision Tree at 90.29%. This suggests that the collection or ensemble of trees did a much better job at handling the multi-class balance across all 7 cover types. The macro recall of the Random Forest was 94.66% which is about 4% higher than the Decision Tree at 90.39%. This suggests that the Random Forest model was significantly less likely to miss minority classes.

The key observation that the baseline Random Forest model outperformed the tuned Decision Tree model, highlights the collective wisdom of hundreds of trees via bagging is more powerful than any single tuned decision tree. By combining hundreds of diverse and slightly weaker trees via majority voting, the ensemble model was able to cancel out individual errors, resulting in a superior accuracy of 95.67%.

Table 3 below presents the best F1 score for each Cover Type in this data set, where the best model for each Cover Type was the Random Forest model.

Table 3. The Best Model and F1 Score for each Cover Type

CoverType	Best Model (F1)	Best F1
Spruce/Fir	RandomForest class_weight=balanced (tuned)	0.9566
Lodgepole Pine	RandomForest class_weight=balanced (tuned)	0.9624
Ponderosa Pine	RandomForest class_weight=balanced (tuned)	0.9514
Cottonwood/Willow	RandomForest class_weight=balanced (tuned)	0.8825
Aspen	RandomForest class_weight=balanced (tuned)	0.8948
Douglas-fir	RandomForest class_weight=balanced (tuned)	0.9227
Krummholz	RandomForest class_weight=balanced (tuned)	0.9593

To examine which variables the Random Forest model relied on heavily to predict the forest cover type, a feature importance chart was generated based upon the mean decrease in impurity. This measures how

often a feature was used to split data across all decision trees and how much those splits reduced the impurity. In other words, how effectively the feature helped separate the data into distinct cover types. Figure 4 presents the top 10 most important features in the Random Forest model, where it is evident that elevation dominates. This is due to the fact that ecologically, tree species have strict altitudinal bands they thrive in due to temperature and climate gradients (Annex A, Figure A.6). Ultimately, this distribution highlights that the model's predictive power is heavily concentrated in a small subset of environmental factors rather than distributed evenly across all 54 features.

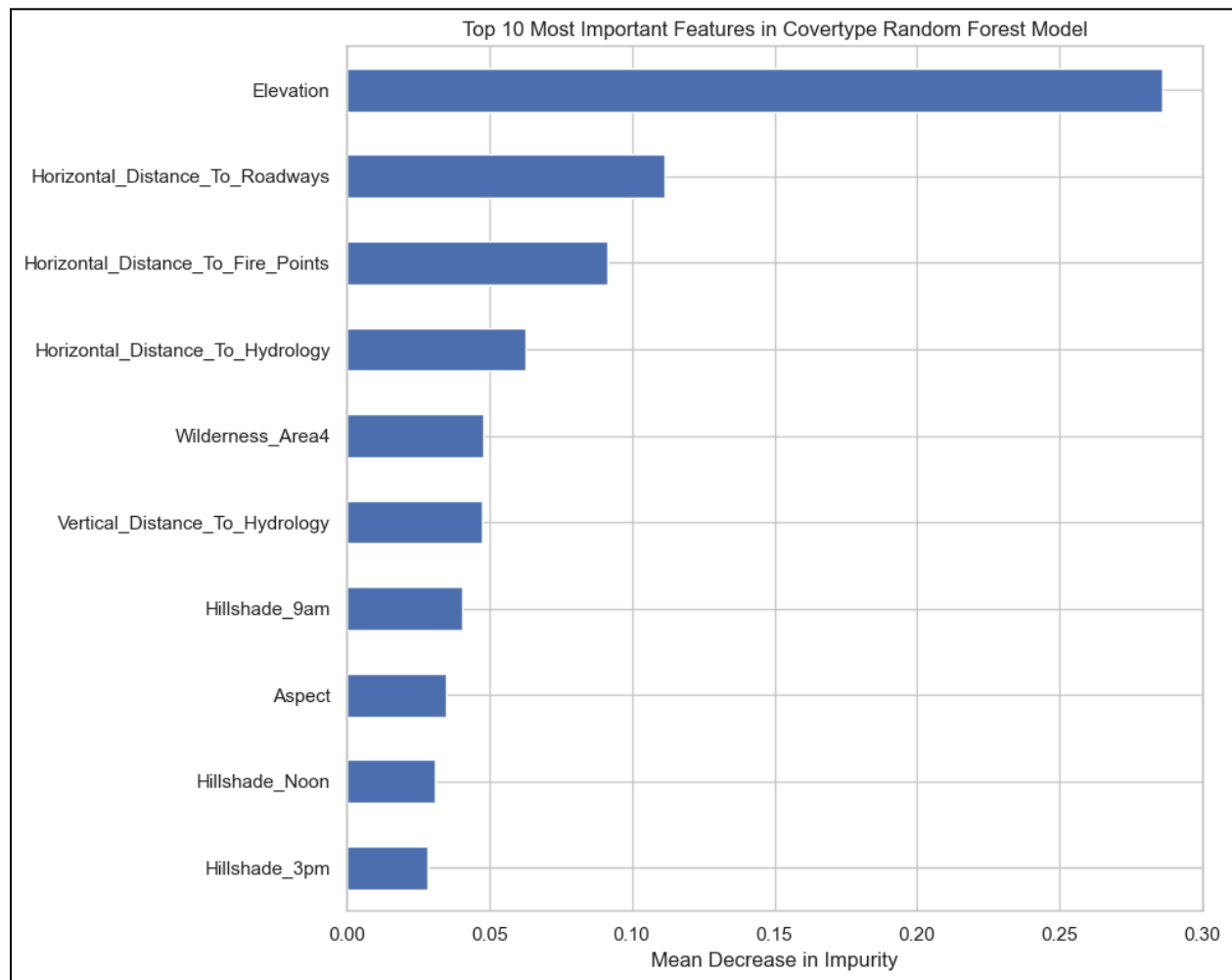


Figure 4. Most important features from final random forest model

6. Conclusion:

Did you prove/disprove your hypothesis or create a useful model?

The primary objective of this project was to compare five supervised machine learning algorithms to determine which model best predicts forest cover type using a combination of environmental and topographic variables. This objective was successfully achieved, with all five models demonstrating the ability to classify forest cover type to varying degrees of accuracy. Among the models evaluated, the **Random Forest classifier** achieved the highest overall performance, producing the strongest Accuracy, Macro Recall, Macro F1-score, and Weighted F1-score on the independent test dataset. These findings

demonstrate that machine learning provides an effective approach for predicting forest cover type and that nonlinear models outperform linear models for this complex multiclass classification problem.

What did you learn about your dataset?

Exploratory data analysis showed that the Forest Cover Type dataset is high quality, containing no missing values and requiring minimal preprocessing before modelling. However, the dataset exhibits substantial class imbalance, with Spruce/Fir and Lodgepole Pine representing the majority of observations while several forest cover types are comparatively rare. The data also showed considerable overlap between some forest cover classes, indicating that the classes are only partially separable based on the available features. Environmental variables such as elevation, soil type, and wilderness area were found to be strong predictors of forest cover type, while the superior performance of the nonlinear models suggests that complex interactions exist among these variables. These findings highlight the importance of using evaluation metrics such as Macro F1-score and Macro Recall, which provide a more balanced assessment of performance across all seven forest cover types.

What would you do next to improve your model?

Several opportunities exist to further improve model performance. Future work could evaluate additional ensemble methods such as XGBoost, LightGBM, or CatBoost, which often achieve excellent performance on structured tabular datasets. Incorporating additional predictors, including climate variables, satellite imagery, or temporal information, may further improve classification accuracy and generalizability. Finally, evaluating the models using data from additional geographic regions or applying spatial cross-validation would provide greater insight into how well the models generalize beyond the Roosevelt National Forest. These improvements could further strengthen the usefulness of machine learning as a decision-support tool for forest inventory, conservation planning, wildfire management, and environmental monitoring.

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Annex A:

Fig A.1 CoverType t-SNE Plot

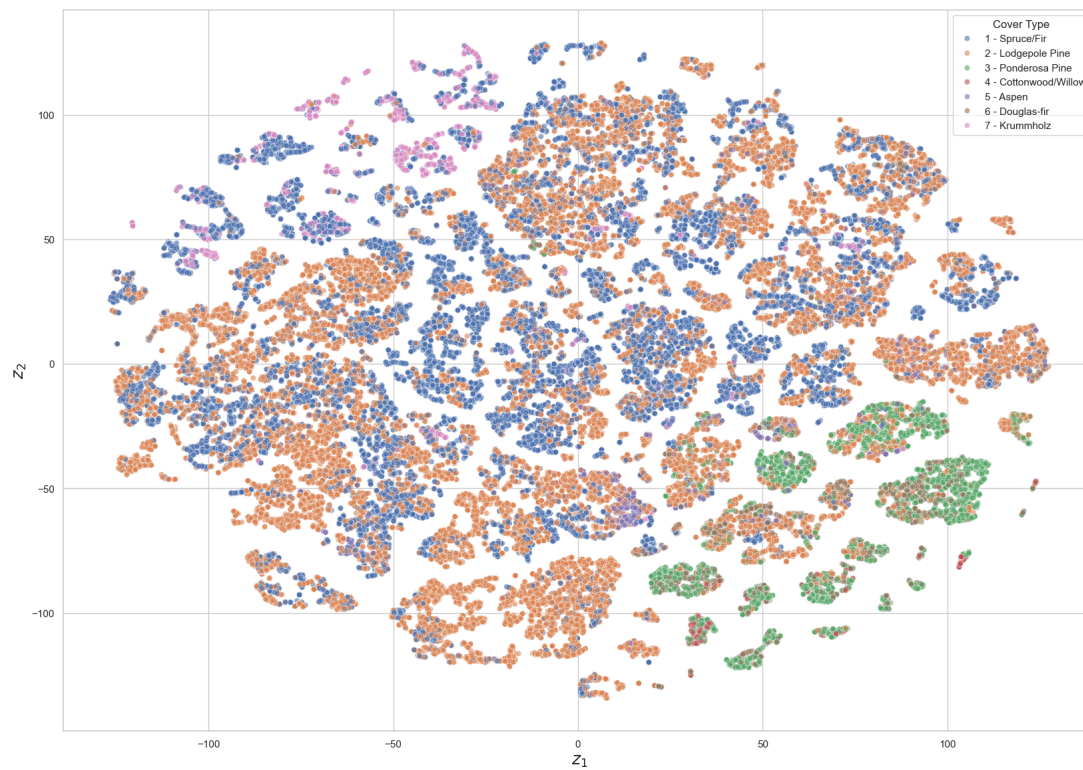


Fig A.2 Cover Type Linear Discriminant Analysis (LDA) Plot

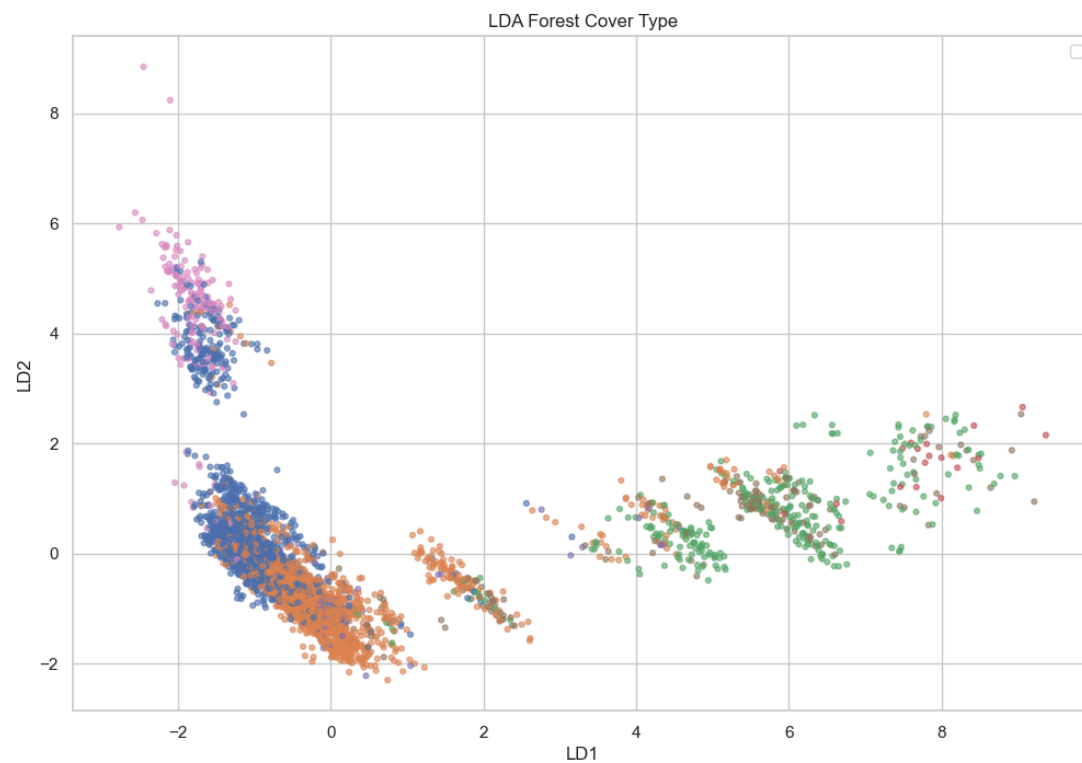


Fig A.5 Neural Net Learning Curves (Accuracy and Loss) Plot

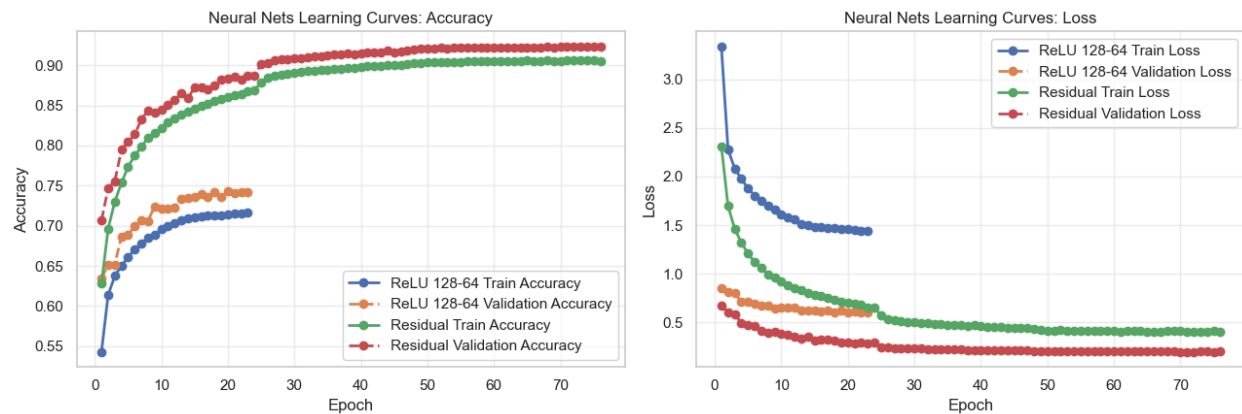


Figure A.6 Elevation Distribution Across Forest Cover Types

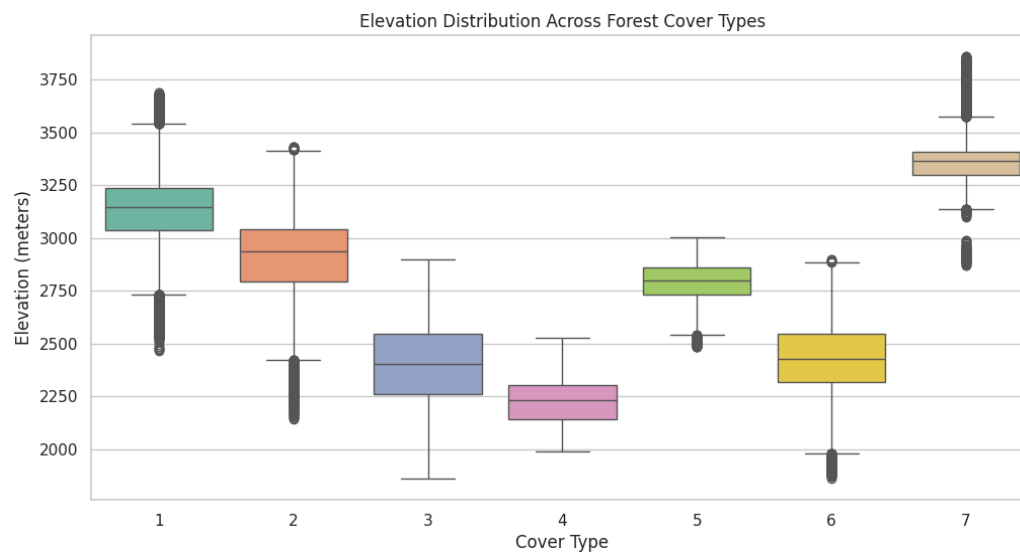


Table A.1: Overview of Classification Models

Model	Scaling	Class Balance	Tuning	Key Parameters
Logistic Regression	StandardScaler	Balanced	None	lbfgs; L2; max_iter=500
Linear SVC	StandardScaler	Balanced	None	Linear kernel
Decision Tree	None	Balanced	Randomized SearchCV	max_depth, criterion, min_samples_split, min_samples_leaf
Random Forest	None	Balanced	Randomized SearchCV	N_estimators, max_depth, max_features, min_samples_leaf
RBF SVC	StandardScaler	Balanced	GridSearch CV	RBF kernel; C & gamma tuned (50k training sample)
Neural Network	StandardScaler	Balanced weights	Manual	Residual network; Adam; BatchNorm; Dropout; EarlyStopping; ReduceLROnPlateau